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A systematic Review on Students' Engagement in Classroom – Indicators, Challenges, and Computational Techniques

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Số tham gia của học sinh là gần 1 triệu thì chỉ có 40% cảm thấy buồn chán, chỉ có 20% HS đặt câu hỏi

ABSTRACT Students' engagement in a classroom is a key factor that influences several educational outcomes. Studies by the University of California, Los Angeles (UCLA) and British universities found that 40% of students are frequently experiencing boredom and less than 20% of students ask questions in class due to poor engagement. A survey by Malaysia's Programme for International Student Assessment (PISA) found that 80% of the participating schools fell into the poor performance bracket. However, studies in this line of research are limited and scattered. To provide a clear insight into this problem and support researchers, we need to understand the current state of research in this area. Consequently, in this paper, a comprehensive review is conducted to map the literature studies to a consistent taxonomy. Search terms revealed 87 papers from several databases that have been classified into seven categories. A systematic review method is applied, analysis is performed, and finally, findings, discussion, and recommendations are presented.

INDEX TERMS Classroom interaction, student engagement; student involvement; student participation;

I. INTRODUCTION

Over two decades, intensive efforts have been made to address persistent educational problems such as low academic performance, high dropout rates, and high rates of student boredom and disaffection (Chapman et al., 2011; Fredricks et al., 2016a). Studies have also reported that students' engagement is a crucial factor in predicting students' academic achievement and influencing several educational outcomes (Salleh et al., 2013). Researchers use several indicators to measure engagement, which includes self-report, attendance rates, teacher ratings, interviews, observations, cross-cultural data and assessments grades (Fredricks & McColskey, 2012; Castro et al., 2015; Reeve, 2004). On the other hand, researchers measure engagement by observing a student's active participation such as the amount of effort and positive emotion or via a student's voice and initiative to take personal responsibility for his/her behavior (Reeve et al., 2004).

In this paper, we review the literature on students' engagement in a classroom. Search terms identified 87 papers from several databases such as Science Direct, Taylor and Francis, Sage, Springer, Wiley Online, IEEE Xplore. The collection of the selected papers is classified into seven categories, overview; dimensions and indications; research studies; problems; factors influence engagement; methods to

measure engagement; and techniques to improve engagement. This paper starts by describing the method used to conduct this review. The method includes a search strategy, search terms, and selection process. We then review the seven categories. We also conduct a comprehensive analysis on the reviewed articles, which include data analysis on students' engagement research, number of selected articles by year of publication, rate of the reviewed articles in different categories, number of articles in different categories by year of publication, and taxonomy of the literature on student engagement. Three potential research problems are discussed, examining the impact of emotion on students' engagement; development of a configuration framework to generate an engagement strategy for a particular classroom environment; utilizing computer-based simulation and virtual environment to study and investigate poor engagement instead of using traditional research tools.

II. METHOD

We first define our search strategy and then describe the selection process to select the most relevant papers in the literature. The specific steps taken during the study's selection process are also described.

A. Search strategy

The electronic databases searched in this review include those identified as relevant to school education, information technology, and social science. In this research, six widely used digital libraries and databases are as follows: Web of Science, SagePub, IEEE Xplore, Springer, ACM Digital Library and Science Direct.

B. Search Terms

The review aims to find all primary research work in conjunction with the terms for possible outcomes. The relevant search terms in the context of this review are identified as follows: “student-engagement in a classroom”; “student engagement in a classroom”; “student-involvement in a classroom”; “student involvement in a classroom”; “student-participation in a classroom”; “student participation in a classroom”. Added to these terms are the keywords, “overview”; “research”; “studies”; “methods”; “status”, “indications”; “problems”.

C. Paper Selection Process

A review search result is further analyzed based on the inclusion and exclusion criteria. It is developed to ensure only relevant works on students’ engagement research are accepted for the review. The following shows the stages of the selection process:

- 1) Search based on the search string.

- 2) Query (as discussed in Section 3.2).
- 3) Excluding duplicates/non-English articles.
- 4) Perform an abstract scan and exclude publications that do not address engagement research.
- 5) Full-text reading to filter and produce final set.

Using these conditions, 87 papers met the inclusion and exclusion criteria and are identified as relevant to the current review. These papers are classified into the following categories: an overview of students’ engagement; dimensions and indications of students’ engagement; studies conducted on students’ engagement research; problems in students’ engagement; factors influence students’ engagement; methods to measure students’ engagement; and techniques to improve students’ engagement.

III. ANALYSIS AND FINDINGS

During the analysis, we found 87 papers on students’ engagement research, nine papers on definition and history of students’ engagement, 12 papers on dimensions of students’ engagement, 13 papers on studies conducted on students’ engagement research, 17 papers on problems in students’ engagement, 10 papers on factors influence students’ engagement, five papers on methods to measure students’ engagement, and 21 papers on techniques to improve students’ engagement. Table 1 shows the detailed result of our analysis.

TABLE I: Data Analysis on Students’ Engagement Research

No	Areas	Reverences
1	Overview of student engagement	Pekrun et al., (2014); Kuh (2009a); Harper, (2009); Coates (2007); Schlechty, (2002); Brewster & Fager, (2000); Bomia et al., (1997); Lamborn et al., (1992); Astin, (1984)
2	Dimensions and Indications of student engagement	Fredricks et al. (2016a); Boekaerts (2016); Kahu (2014); Salleh, et al., (2013); Finn and Zimmer (2012); Linnenbrink-Garcia et al. (2011); Trowler (2010); Eccles (2009); Luo et al., (2009); Appleton et al. (2008); Fredricks et al. (2004); Finn (1989)
3	Studies conducted on student engagement research	Salim & Ayub (2017); Krauss et al. (2017); Ayub et al. (2017); Fredricks et al. (2016b); Wang et al. (2016); Ishak, & Amjah (2015); Martin et al. (2015); Teoh et al. (2013); PISA survey (2012); Aiedah & Audrey (2012); Willms et al., (2009); Akey (2006); Learning Point Associates (2004)
4	Problems in student engagement	Lawson& Silver (2013); Nikish (2013); Mustapha & Rahman (2011); Pryor et al. (2010); Kazmi (2010); Macal & North (2009); Thang (2009); Mann & Robinson (2009); Dasari (2009); Sayadi (2007); Ming & Alias (2007); Yazzie-Mintz (2007);Bousquet and Le Page (2004); Stipek & Byler (2004); McInnis (2001); Marks (2000); Astin & Sax (1998)
5	Factors influence student engagement	Wubbels at al. (2016); Ricard & Pelletier (2016); Shernoff at al. (2016); Abdullah at al. (2012); Christenson at al. (2012); Lam at al., (2012); Barkatsas et al. (2009); Carroll et al. (2009); Skinner et al. (2008); Kaplan et al. (1997)
6	Methods to measure student engagement	Castro et al. (2015); Veiga at al. (2014);Fredricks & McColskey (2012); Parsons & Taylor (2011); Reeve at al. (2004)
7	Techniques to improve student engagement	Goyal and Krishnamurthy (2018); Falcão et al. (2018); Wang (2017); Heflin et al. (2017); Ding et al. (2017); Chen and Chiu (2016); Russell et al. (2016); Imlawi et al. (2015); Tizzard (2010); Maheady & Gard (2010); Zepke & Leach (2010); Dunleavy and Milton (2009); Up (2009); Frondeville (2009); Rathvon (2008); Krause & Coates (2008); Barnes et al. (2007); Taylor et al. (2006); Windham (2005); Kvavik et al. (2004); Marzano at al. (2003); Dim, Kim & Orey (2017)

Figure 1 shows the number of reviewed articles in the seven categories according to the years of publication. The distribution from 1984 to 2018 is shown. The figure shows that, according to the reviewed articles, a few studies have been conducted between 1984 and 2003 at a rate of 12% of total reviewed papers. In the next five years, the rate increased to 21%. In the last decade, the rate reached 67%. This clearly shows that research in students' engagement is receiving increasing attention.

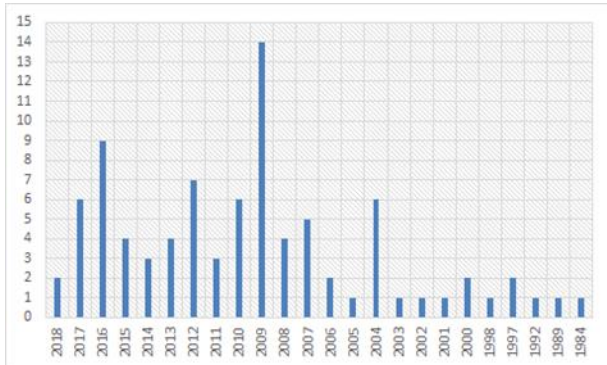


Figure 1 Number of selected articles by year of publication

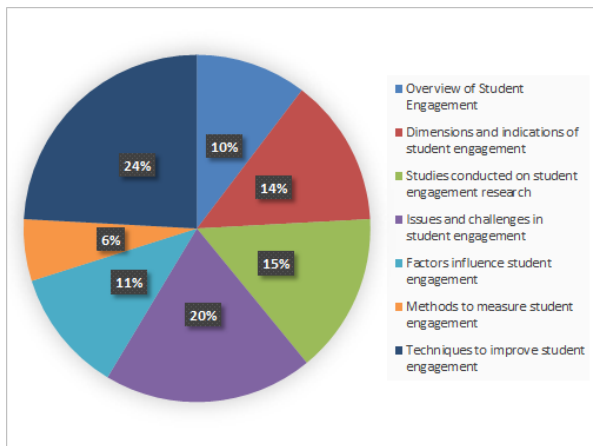


Figure 2 Rate of the reviewed articles in different categories

Figure 3 shows the identity of the source databases of the reviewed articles. Most of the selected articles are from Science Direct, Taylor and Francis, Sage, Springer, Wiley Online, IEEE Xplore, and IGI global, which constitute about 72% of the selected articles. Some other supporting articles have been selected from other databases.

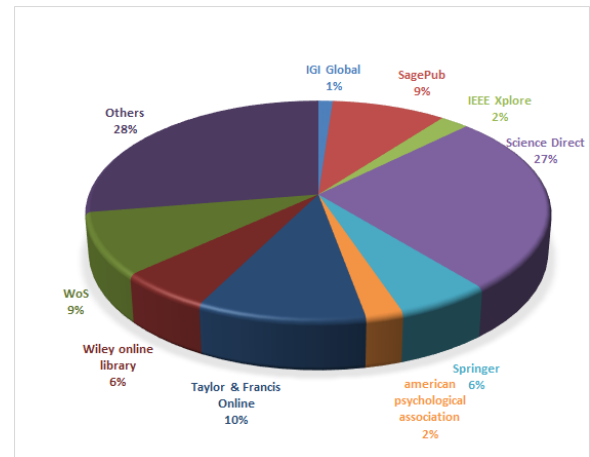


Figure 3 Identity of the source databases of the reviewed articles

Figure 4 shows the number of articles in each category and according to the year of publication. Again, the results show that most of the recently published articles are about the category “techniques to improve student engagement”, which clearly reflects that the problem of poor engagement is gaining more attention. The research in “Studies conducted on students' engagement” and “factors influence students' engagement” are also high. Generally, the figure shows that research in students' engagement is rapidly increasing.

Figure 5 shows the taxonomy of literature on students' engagement. The first category is the “overview”, in which two topics are discussed under this category, definitions, and history. The results show that the literature on students' engagement can be traced back to at least 70 years ago and it received more attention from researchers since the mid-1980s. The second category is the dimensions and indications. Four dimensions have been identified which are behavioral engagement, emotional engagement, cognitive engagement, and social engagement, each of which has a number of indications. The results show that emotional engagement received less attention while social engagement is just recently discussed in the literature and it requires more research.

The third category is “studies conducted on students' engagement”. The results show that there are a number of studies that have been made either on the subject domain such as mathematics and science subjects or on the level domain such as primary or secondary school students. The results also show that there is a broad agreement in the literature that students' engagement is a multi-dimensional construct in either subject domain or level domain; the most prevalent dimensions are, behavioral, emotional/ affective, and cognitive engagement. Among these, behavioral engagement has received the most attention and emotional engagement, the least. Another issue is that tertiary level institutions are facing more signs of disengagement or lack of commitment to studies. The fourth category is “Problems in students' engagement”, the issues and challenges of which have been presented. The results show that the issue of poor engagement

often arises within middle school and aggravates during high school and becomes worse at the university level. Several issues have been revealed such as students are not motivated; unable to manage their study workloads; lack of personal autonomy; teacher-centered learning process; low-level class participation; bored in class; and lack of interactions with teachers.

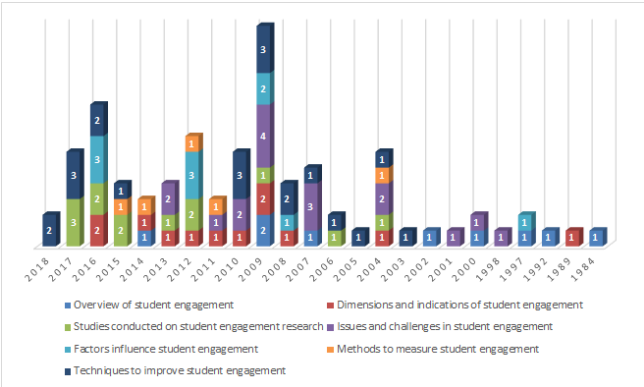


Figure 4 Number of articles in different categories by year of publication

The review also reveals that one of the main challenges to study these issues is the limitation of traditional research tools. Studies indicate that traditional methods consume time and effort, difficult to manage and control human subjects to conduct experiments; and experiments applied to real environment (classroom) have limited settings to be tested and the costs would be excessive if multiple settings are applied.

Under the fifth category “Factors influencing student engagement level”, seven factors have been identified. The results show that teachers have a significant role among the relative factors postulated to influence students’ engagement such as parents, friends, students’ characteristics and learning environment. A teacher’s role could be classified into a teacher’s control, teacher’s support, and teacher-students’ interactions. The sixth category is “Methods to measure student engagement”, the results of which show that there are three indicators that could be utilized to measure or evaluate students’ engagement which are participation level, motivation level, and their positive or negative behavior.

The last category is “Techniques to improve students’ engagement”. Several technologies have been deployed to enhance engagement and subsequently, educational outcomes. The results show positive attempts by researchers to engage students using multimedia tools, mobile technology, and web-based systems. However, these techniques might not be applicable in real situations due to the absence of one or more implementation requirements such as limitation of time, resources, teacher capability, and suitable environment.

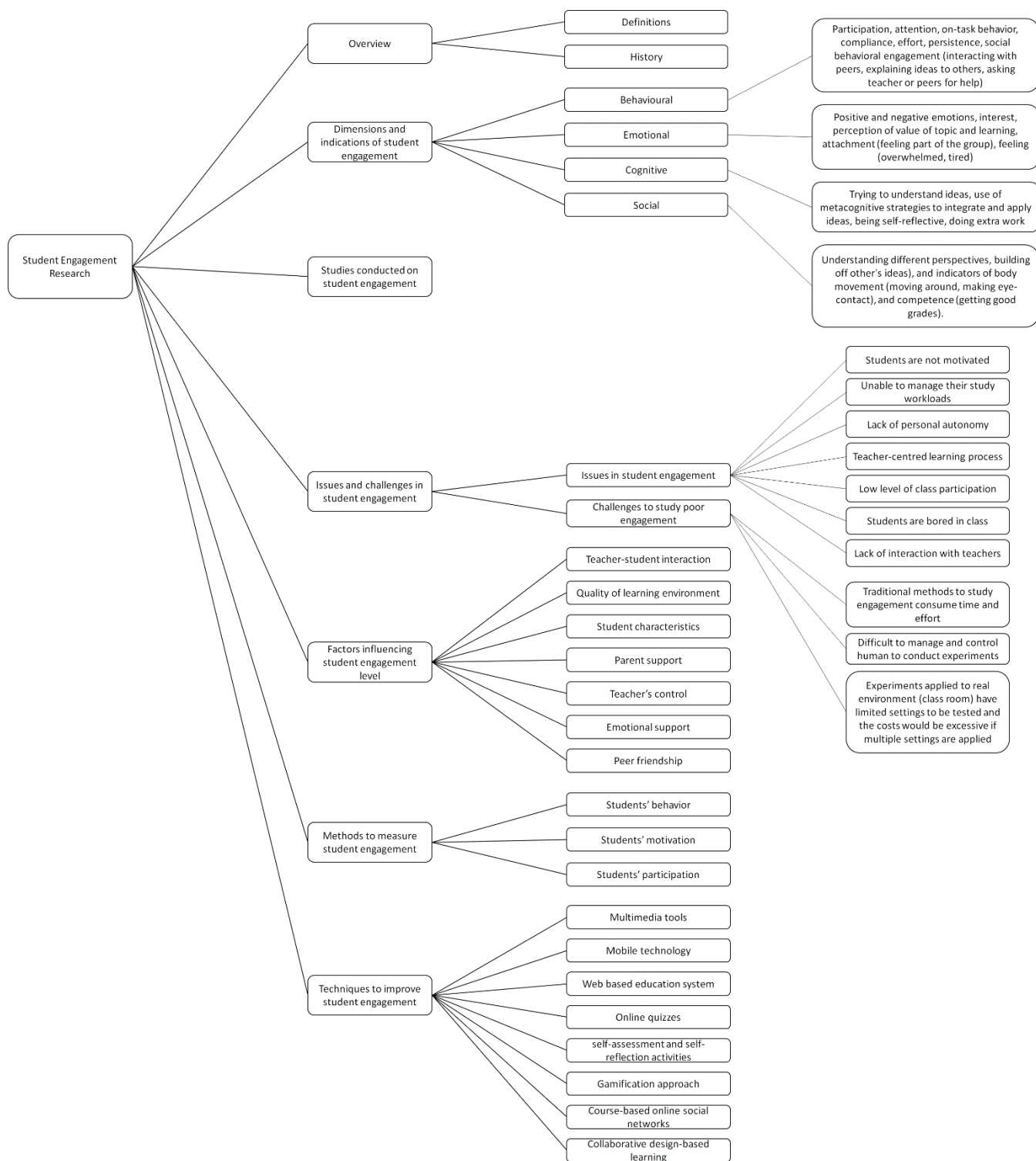


Figure 5 Taxonomy of literature on students' engagement

IV. STUDENT ENGAGEMENT RESEARCH

Students who are engaged receive more teacher involvement, highly motivated to attend classes, and actively involved in classroom discussions (Skinner & Belmont, 1993). Disengaged students are more likely to find that teachers become more controlling and increasingly withdraw their

support, thereafter develop habitual truant and/or occasionally misbehave in classrooms (Skinner & Belmont, 1993). Such students end up getting poor academic results, involving in many disciplinary actions and occasionally dropping out of school or colleges.

A. Overview of Student Engagement

Based on Trowler (2010), students' engagement refers to the interaction between the time, effort and other related resources dedicated by both students and their institutions intended to optimize the student experience, knowledge and improve the academic performance, and reputation of the institution". Kuh (2009a) define student engagement as "participation in educationally effective practices that lead to a range of measurable outcomes". In other words, student engagement can be examined by observing students' level of participation in classroom tasks or activities, their interest, enthusiasm, and care level shown in completing the task given successfully. Besides, student engagement also promotes both emotional wellbeing and learning outcomes. Research has highlighted the negative effect boredom can have on emotional wellbeing on students (Pekrun et al., 2014). In Student Engagement Literature Review by Trowler (2010), he states this research gained widespread attention particularly in North America and Australasia where the large scale of student engagement survey took place. Most prolific researchers such as Kuh (2009b) and Coates (2007) developed and implemented large scale of national student engagement surveys within several universities.

According to Kuh (2009a), a theory of students' engagement has been deliberated in the literature for more than 70 years and attracted widespread attention of social researchers in the literature since the mid-1980s. Its beginning can be traced back three decades ago via Alexander Astin's work on student involvement in 1984 (Astin, 1984; Trowler, 2010). Primarily, student engagement has been focused upon improving achievement, positive behaviors, and a sense of belonging in students so that they remain in school. Progressively, student engagement strategies are further developed and broadly applied as a way to manage classroom behaviors. Harper (2009) postulated engagement is more important than involvement or participation as it requires feelings and sense-making as well as activity. The past twenty years of research on students' engagement indicate that engagement and motivation lead to higher achievements of the students' performance (Brewster & Fager, 2000). In a nutshell, for a good purpose, a huge number of researchers have studied and investigated students' engagement due to deteriorating engagement over the years (Schlechty, 2002).

B. Dimensions and Indications of Student Engagement

Initially, Finn (1989) introduced two-component student engagement models; Participation and Identification (PI) Model. In his terms, Participation refers to "behavioral engagement" and Identification refers to "affective/emotional engagement". Since many researchers have shown interest to pursue engagement research, the multi-dimensional model becomes more popular. Therefore, Fredricks et al. (2004) suggest that students' engagement is a multidimensional construct composed of behavioral, emotional, and cognitive

components. They discuss the potential of students' engagement via motivation, belonging, and school climate.

The importance of students' engagement is emphasized in educational institutions as it is observed by researchers that many students are bored, unmotivated, uninvolved, and disengaged from the academic and social aspects of school life (Marks, 2000). These subtypes are foundational to understanding the engagement issue. Figure 6 shows the dimensions of Student Engagement.

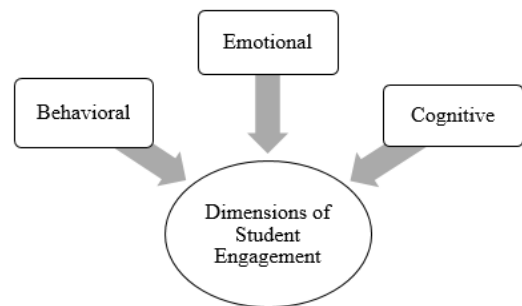


Figure 6. Dimensions of Student Engagement

- i. The *Behavioral* dimension concerns students' adhere to the classroom and school rules, for instance, their class attendance and politeness as well as students' involvement in classroom work and discussions and in extracurricular activities.
- ii. The *Emotional* dimension is reflected by students' own feelings, attitudes, and opinions toward school. In other words, it mainly refers to liking school, belongingness, and interests for learning.
- iii. The *Cognitive* dimension concerns students' involvement in learning, for example, willingness to engage in learning, and task-oriented goals

However, recently, some researchers have added a social dimension to these multidimensional of engagement due to the increasingly vital role of social interactions in learning process (Fredricks, Wang, Linn, Hofkens, Sung, Parr, & Allerton, 2016). For instance, a study by Linnenbrink-Garcia, Rogat, and Koskey (2011), they included a social dimension of engagement, where they define "Social forms of engagement involved around classroom tasks and assignments, including good participation among classmates and the quality of social interactions". Similarly, Finn and Zimmer (2012) defined social engagement as "Students' prosocial behavior in classrooms and the quality of interactions with peers around instructional content".

In another research, teachers' and students' indicators of engagement are identified in interview sessions and implied in the three dimensions of engagement (Boekaerts, 2016). Fredricks et al., (2004) postulated that the design of engagement across these three dimensions have a long-term impact on students' successful academic result and hypothesized the need to assess the distinguished role of these three dimensions of engagement. According to Luo, Hughes, Liew, & Kwok, (2009), if students emotionally disengaged

during a lesson, they tend to increase school absences and decrease efforts and attention on their academic subjects, both of which eventually lead to declines in academic performance and a reduced interest in attending classes. Eccles (2009) posited that emotional belonging to school dimension serves a more essential role in terms of motivating students to strive for excellence. Students experience positive feelings of belonging to a school, as they enjoy interacting with their classmates and lecturers. However, if students do not feel belonging to school, it may not motivate them and enhance their academic performance.

Based on Appleton, Christenson, & Furlong, (2008) research, the critical role of both behavioral and emotional/affective components account for high rates of student engagement. In another research, Salleh, et al., (2013) conducted a study to examine Malaysian students' school engagement status and to understand the factors that influenced their engagement in three psychological domains (cognitive, emotion, and behavior). Their results show that the level of students' engagement in school based on the domain of emotion is the highest, followed by cognitive and behavior.

Furthermore, Kahu (2014) focused on two key elements of emotional engagement, which are interest and belonging. His findings highlight the importance of interest triggered by personal preferences and experiences. Interest led to enjoyment, increased behavioral engagement with greater time and effort, and improved cognitive engagement in terms of depth and breadth of learning. Moreover, Trowler (2010) propose that "each of these dimensions can have both a 'positive' and a 'negative' pole, each of which represents a form of engagement, separated by a gulf of non-engagement (withdrawal, or apathy)". He reported students either engage positively or negatively in behavioral, emotional or cognitive dimensions.

C. Studies Conducted on Students' Engagement Research

Previous studies have shown that levels of engagement are directly related to academic achievement in some particular subjects (Finn, 1989; Fredricks et al., 2004; Akey, 2006). Therefore, there are few studies that aim to investigate subject-specific students' engagement in Mathematics, Science, Malaysian Studies and English Language. A study conducted by Martin et al. (2015) reports that engagement in mathematics and science subjects deteriorates during the secondary school years. Hence, Wang et al. (2016) develop and validate student and teacher-report survey to measure students' engagement in mathematics and science by proposing a bifactor modeling approach. Their bifactor model suggests that mathematics and science engagement is characterized by multi-dimensional: behavioral, cognitive, emotional, and social engagement.

According to Wang et al. (2016), "A multi-dimensional perspective on students' engagement provides a richer characterization of how students act, feel, think, and socialize

with others in mathematics and science classrooms, rather than considering each of the dimensions separately". Their findings confirm that students' engagement is comprised of multiple dimensions with proof to support a bifactor structural model. Similar to Wang et al. (2016), another research by Fredricks et al. (2016b) conduct in-depth interviews with 106 students and 34 middle and high school teachers about how they conceptualized mathematics and science engagement and disengagement. Their qualitative analysis of student and teacher interviews support the multi-dimensional construct of engagement (Fredricks et al., 2016b). They also develop and validate a new student self-report to measure mathematics and science engagement.

In another study, Ayub et al. (2017) investigate mathematics engagement by focusing on the cognitive, affective and behavioral engagement domains among secondary school students. A total number of 387 students from urban and rural secondary schools in Pahang, Malaysia, were randomly selected. According to PISA survey conducted in 2012, engagement performance in mathematics lesson declined among Malaysian students. Thus, in their research, they suggest that teachers should restructure their teaching strategy by creatively devising new techniques in teaching mathematics. Consequently, when planning for mathematics activities or strategies, crucial elements of engagement; cognitive, behavioral and emotional are accounted for to improve engagement in mathematics lessons. The researchers also emphasize strategies that are needed to be integrated during the teaching should not be too teacher-centered (Ayub et al., 2017).

In a similar study conducted at a public university, Universiti Putra Malaysia, the researchers investigate the relationship between students' attitudes toward statistics and mathematics engagement among 293 undergraduate students who are enrolled in statistics courses from several programs (Salim & Ayub, 2017). Their analysis of mathematics and statistics engagement domain reveals that attitudes towards statistics are positively related to the emotional domain, then a cognitive domain and followed by a behavioural domain. Their findings show attitudes are vital for students to be engaged in mathematics and statistics (Salim & Ayub, 2017).

In the Malaysian context, lecturers face challenges in teaching theoretical subjects like Malaysian Studies and English Language because students complain that these subjects are irrelevant and not engaging (Aiedah & Audrey, 2012). Therefore, the researchers propose the application of project-based learning to minimize the gap between the theoretical knowledge and the relevancy of subjects. Through their study, it is found that project-based learning is an approach to improve students' engagement specifically in four aspects such as responsible learning, strategic learning, collaborative learning, and energized by learning (Aiedah & Audrey, 2012; Learning Point Associates, 2004). However, a lack of commitment from students and time-consuming are the barriers to implement project-based learning. They suggest

that the lecturers can alleviate the difficulties in practicing project-based learning if they are well-equipped with the knowledge and skills in executing this strategy. In another study by Willms et al. (2009), two social studies teachers and 24 secondary school students are involved. From their analysis, they found that students display engagement as they engage behaviorally and emotionally and enjoy their assessment in a group.

Ishak and Amjah (2015) state “Positive body language, consistency in focus, verbal participation, student confidence, fun, and excitement play deterministic roles in conceptualizing students’ level of engagement”. From their observations, they found that students’ level of engagement for the first activity is neither high nor low. This is apparent in their active participation and their exhibition of interest and enthusiasm in group work. This study is beneficial to teachers to be aware of the importance of engaging students in learning, especially, in Social Studies (Ishak & Amjah, 2015). Another study investigates the characteristics of students’ engagement in a local public university. Teoh et al. (2013) distribute the College Student Experience Questionnaire (CSEQ) to a total of 64 first and second-year students. The findings reveal five indices of students’ engagement. Students rate highest on “life-long learning” index, followed by the indices of “active learning”, “cooperation among students”, “experience with diversity” and lastly, “student-faculty interaction” index. Further investigation reports other aspects such as imaginative teaching practice, prompt feedback to students’ work, react fairly to students’ performance and suitable teaching strategy would encourage involvement and commitment of students in their studies.

Even though there is a broad agreement in the literature that students’ engagement is a multi-dimensional construct and some notable research use variations ranging from two to four dimension; behavioral, cognitive, emotional and social. Despite these inconsistencies, the most prevalent dimension in the literature is that engagement consists of three distinct dimensions; behavioral, emotional/affective and cognitive engagement (Fredricks et al., 2004). Among these, behavioral engagement receives the most attention and emotional engagement, the least (Fredricks et al., 2004). From our analysis, a few prominent research use this crucial multi-dimensional focus in subject domains or among primary/secondary school students. However, past investigations report that tertiary level institutions face more signs of poor engagement or lack of commitment to studies (Kazmi, 2010). Furthermore, Krauss et al., (2017) report that existing studies on school engagement have been mostly limited to individual and school-based predictors in Western countries.

D. Problems in Students’ Engagement

Issues of poor engagement often arise within middle school and are aggravated during high school (Krauss et al., 2017). Marks (2000) and Stipek and Byler (2004) report that there is

a steady decline from the middle to high school students in affective interest and motivation to learn, followed by a decline in cognitive enthusiasm to face challenges in tasks. Furthermore, evidence supported by Willms et al. (2009) shows that poor engagement usually becomes a concern in middle school and high school.

Studies conducted in Australia and the United States highlighted that the engagement performance in higher education is declining as undergraduates are becoming less involved with university activities and not on par as the previous generation. McInnis (2001) also stated that students are not motivated to study and unable to manage their study workloads due to improper time management. Besides, Astin & Sax (1998) observed that students spend less time seeking teachers’ advice, oversleeping and absence from classes. They also become materialistic and expect university to assist them in achieving instrumental goals.

Despite the widespread studies of students’ engagement in the higher learning institutions of Australia, USA, and Canada (Kazmi, 2010), students’ engagement research in Malaysian public universities are scant (Teoh et al., 2013). Thang (2009) and Ming and Alias (2007) report that a majority of students in public and private universities of Malaysia experience basically a teacher-centered learning process and lack personal autonomy. Based on Dasari (2009) research, he found that Asian students are notorious for their low-level class participation. Furthermore, Sayadi (2007) reveals that less than 20% of students ask questions to the lecturers during class. This is due to their disinterest or boredom with on-going lessons and is not concerned with participating in in-class activities. Mustapha and Rahman (2011) further define the term “negatively passive participation” as quiet; not concern about class activities; not interested in the lessons and not paying attention to class.

Studies by UCLA Higher Education Research Institute and British universities found that 40% of students are frequently bored in class due to poor engagement in classrooms (Pryor, Hurtado, DeAngelo, Blake & Tran, 2010) and 59% found lectures are boring in at least half of their classes (Mann & Robinson, 2009). Similarly, an annual survey by Indiana University’s High School, discovered that about 30% of the students indicate that they are bored due to the lack of interaction with teachers and 75% report that the subjects being taught are not interesting (Yazzie-Mintz, 2007; Nikish, 2013). Besides that, 34% of students said that a primary source of their boredom was “No interaction with teacher”.

On the other hand, many researchers study the problem of poor engagement in classrooms using traditional methods such as the questionnaire survey and setting up experiments that consume considerable time, effort and with very limited settings. If they need to repeat an experiment with different settings, gathering a huge crowd of students would be tedious. Furthermore, since these experiments are applied to humans, there are very limited settings to be tested and the costs would be excessive if multiple settings are applied (Macal & North,

2009). Studies have indicated that it is challenging to manage and control humans to conduct multiple experiments or repeat experiments with the same settings (Bousquet & Le Page, 2004). Thus, traditional methods have a poor success rate in mitigating the problem. In conclusion, according to Lawson and Silver, (2013) and Teoh et al. (2013), students' engagement occurs when there is a reaction by students to the lecturer's teaching and learning process in an active learning environment.

E. Factors Influencing Students' Engagement Level

Apparently, high-level engagements in classrooms are related to enhanced achievement, attained knowledge, skill and effective learning (Barkatsas et al., 2009). Better engagement produces better emotional functioning (Skinner et al., 2008). The purpose of engagement in the classroom is to serve as a protector against student drop out issues and to prevent students' involvement in unhealthy activities (Skinner et al., 2008). Research findings indicate that students' engagement is correlated to academic performance, and their disengagement leads to poor academic performance in a variety of subjects or domains (Coates, 2007).

Recent results reveal that parent support for basic psychological needs seems to be the most significant factor in either academic motivation or dropping out cases (Ricard & Pelletier, 2016). In addition, parental monitoring and family cohesion are reported as predictors of engagement (Krauss et al., 2017). Further findings indicate that supportive relationships with lecturers are significantly associated with students' engagement (Krauss et al., 2017). According to Abdullah et al. (2012), students' personality, environment, and influence of instructors and peers are factors that are able to motivate students' engagement throughout the lesson. Ricard and Pelletier (2016) also agree that peer friendship affects motivation, persistence, and engagement in a classroom.

In another research, Shernoff et al. (2016) use the Experience Sampling Method (ESM) to predict students' engagement and experience. Their demonstration shows that the quality of the learning environment contributes to students' engagement and a sense of classroom self-esteem. Even though students' characteristics substantively influence their motivation, learning, and quality of learning environment directly or indirectly influence the students' academic performance (Wubbels et al., 2016). It is also believed that students' engagement in the classroom is influenced by teachers' control (Wubbels et al., 2016). Teachers' role is essential for the quality of learning process and students' experiences (Christenson et al., 2012). In addition, teachers' support and caring have been found to be pivotal in students' engagement. Carroll et al. (2009), state that the teacher is the strongest factor after students' characteristics. The teacher influences the students via a variety of activities such as assisting students, providing instructional, emotional and organizational support. Based on the findings by Lam et al. (2012), among the relative factors postulated to influence

students' engagement, teacher-student interaction has received significant endorsement for being the resilient predictor of engagement and the most substantial contributor to academic outcomes. Figure 7 depicts the factors that influence students' engagement.

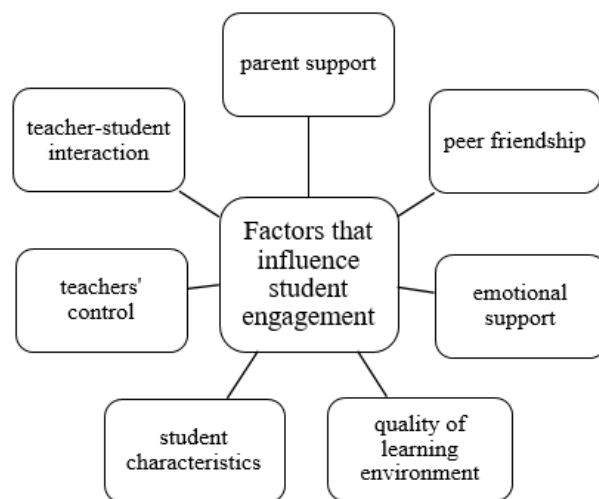


Figure 7. Factors That Influence Student Engagement

F. Methods to Measure Students' Engagement

Another approach to defining student engagement is to see how it is measured. For centuries, few common measures have been practiced to identify if students are actively engaged or involved in the learning process. These measures mainly focused on the traditionally "quantifiable" aspects of student behavior, attendance rates, truancy, time-on-task, and consequently suspension/discipline rates (Parsons & Taylor, 2011). Generally, highly motivated student always attend the classes and score standardized test. However, demotivated students always absence to class and possess poor academic performance. Researchers stressed that more systematic and thoughtful attention to the measurement of engagement is the most imperative direction for future research. (Fredricks & McColskey, 2012; Veiga, Reeve, Wentzel, & Robu, 2014).

Recently, Fredricks and McColskey (2012) reviewed the literature to identify instruments that have been used to assess student engagement in the upper elementary to high school years. In the review, they found a limited number of self-report engagement measures that included scales to assess all three of the dimensions. Moreover, the items in these instruments were used inconsistently across behavioral, emotional/affective, and cognitive engagement. There was also limited evidence to support the validity of several of these measures. Another concern was that the majority of measures identified in the review focused on general engagement in school rather than on engagement in specific subject areas (Fredricks & McColskey, 2012). They prioritize more on school rather than on tertiary institutions.

Other indicators that have been used to measure engagement in the literature include teacher ratings,

interviews, observations, cross-cultural data and assessments grades (Fredricks, & McColskey, 2012; Castro et al., 2015; Reeve et al., 2004). However, the effective disposition of lecturers and students has not been comprehensively studied to understand its effects on students' engagement in classrooms. On the other hand, extra guidance and culture of explanation instead of the culture of right answer are other ways to measure students' engagement. Similarly, as reported by Fredricks & McColskey (2012), students are encouraged to participate in a classroom by honestly self-reflecting their understanding of the lesson before moving onto the next lesson. For example, each student can rate their understanding from 1-3 (1 is lowest understanding rate and 3 is highest understanding rate). By rating their understanding, they can reflect what was taught.

An impressive way to measure students' engagement is to ask questions in a classroom. Typically, when a question is asked, the same person will raise each time. This issue might lead to inattention in the classroom. Thus, overall students' participation is required to measure the students' engagement (Trowler, 2010). In another research, they reveal that students who shared a good interaction with their lecturer reported higher levels of emotional students' engagement compared to those who shared poor interactions with their lecturers. Improved quality of work is a sign of good engagement in a classroom (Trowler, 2010). Generally, from our analysis, we found that students' behavior, motivation, and participation infer the engagement level of the students in the classroom.

G. Computational Methods/Techniques to Improve Students' Engagement

For humans, the process of evaluating students' engagement is simple and straightforward. It can be done by monitoring students' misbehavior, motivation, and participation. However, to improve the intensity of engagement is very challenging. While different lecturers have different teaching strategies, but how effective are these strategies? Which strategy is more efficient? Is the effectiveness of a particular strategy depends on situations? Are there different situations and suitable plans for each situation or group of situations? Many questions have not been clearly answered by the literature due to the difficulty of conducting experiments on humans. When teachers lack the resources, ideas, and materials to effectively manage the social and emotional challenges within the particular context of a classroom, students show poor interest in on-task behavior and performance (Marzano, Marzano, & Pickering, 2003). Therefore, many researchers proposed methods or techniques to engage students with lessons despite the factors influencing it. Table 2 shows examples of techniques in the literature.

Windham (2005) recommends the concept of "interaction, exploration, relevancy, multimedia and instruction" for students' engagement activities that strengthen the interaction of lecturers with their students and at the same time monitor how their students learn. Students are given the opportunity to

interact with someone successful in their field as opposed to being theoretical and text-based. Applying real-life scenarios focused on keeping young students engaged and motivated. Engagement is a major condition for learning, in fact, outdated educational systems fail to sustain the interest and objectives of millennials generation which is infused with digital technologies (Falcão et al., 2018). Innovative and fun context of teaching strategy is essential to promote engagement. A variety of multimedia tools and incorporating technology into learning activities have been shown to increase students' interest, higher achievement, improved quality of work, higher students' motivation, and improved information literacy and critical thinking skills (Taylor et al., 2006). For example, using mobile devices for research and social-based learning activities have been shown to upsurge their engagement by as much as 78% (Dunleavy & Milton, 2009).

Heflin et al. (2017) investigate the impact of mobile technology on students' attitudes, engagement, and learning. Their findings indicate that mobile technology is allied with positive students' perceptions of collaborative learning. This clearly shows that multimedia and technology have been proven to be helpful in engaging students, exploring ways to present their ideas (Dunleavy & Milton, 2009; Barnes et al., 2007). Kvavik et al. (2004) state, "Students recognize a number of benefits of classroom IT, which include the convenience, management of classroom activities, time savings, improved learning, better and more effective communications and better presentation of their class assignments".

Recently, web-based education systems have been a popular topic. Goyal and Krishnamurthy (2018) suggest that cognitive strategies have a vital role to design web-based systems. Students utilize strategies when learning any subject from online learning materials. The purpose of a web-based education system is to better organize the content that caters to different characteristics of students in the learning process. In recent research by Russell et al. (2016), they report that the use of learning materials and formative assessments using online quizzes are helpful in their learning and students agree that the course is relevant to real-world applications and to their daily life. It also increases students' participation rates, achievement, and on-task behavior.

In a study by Wang (2017), she identifies that a course design improves students' behavior, emotional and cognitive engagements that promote high achievement in their studies. In addition, her findings indicate that engagement in self-assessment and self-reflection activities have an important impact on online studies and social interaction, which produce a significant engagement in problem-solving activities (Wang, 2017). Ding, Kim, and Orey (2017) emphasize on gamification approach to foster students' engagement. They developed a gamified online discussion tool, gEchoLu to examine students' engagement in online discussions. Their results indicate that gEchoLu has positive effects on student

behavioral, emotional, and cognitive engagement (Ding et al., 2017).

In another study, Imlawi et al. (2015) analyze the impact of students' engagement in course-based social networks and observe educational outcomes. Their findings recommend that teachers who create online course-based social networks to interconnect with their students are able to improve their engagement, motivation, and satisfaction. In addition, they provide strategies for teachers to fit their activities and enhance their students' engagement and educational outcomes.

In addition, Up (2009) indicates that technology improves students' engagement including cognitive, affective,

behavioral, academic, and social engagement. Therefore, lecturers need to find techniques to motivate their students to learn and increase participation rates so that their time is devoted to learning and there is no room for misbehavior. Despite the advantages of digital technology in promoting students' engagement, Chen and Chiu (2016) propose and develop an intergroup competition mechanism and integrate it into a multi-touch platform for collaborative design-based learning (DBL). Their analyses show that students who participate in intergroup competition significantly foster better students' engagement, learning achievement, and creativity.

Table 2. Computational Methods / Techniques to Engage Students

Reference	Techniques
Zepke & Leach (2010)	- Enhance students' self-belief, work autonomously, enjoy learning with others, and competent to achieve their own objectives. - Create learning that is active, collaborative, and fosters learning relationships. - Enable students to develop their social and cultural capital. - Invest in a variety of support services. - Adapt to changing student expectations. - Enable students to become active citizens and develop their social and cultural capital.
Krause & Coates (2008)	- Stimulate discussion and debate, exploration and discovery. - Stimulate question answer sessions and a range of interactive activities. - Provide feedback and continuous assessment tasks early and often. - Engage students in self-assessment and peer assessment. - Engage students socially by asking for their opinions and acting on them. - Engage students emotionally by providing examples of career success.
Tizzard (2010)	- Use of free software such as Xtranormal to create videos as learning objects, weekly reminders, and FAQ videos. - Interactive chat room experience. - On-line quizzes as self-assessment tools.
Parsons & Taylor (2011)	- Use technology in the classroom. - Reduce lectures, increase group-discussion time, and ignore the "multitasking" student.
Windham (2005)	- Interaction – Connect with experts and expertise - Exploration – hands-on, inquiry-based, problem-based, and exploratory, new forms of digital media (video games, social media, and websites) - Relevancy – Apply real-life scenarios - Multimedia – multimedia tool (e.g. WebQuests, blogs, wikis, YouTube, video documentaries), iPhones and other mobile devices for research. - Cameras, video, and video editing, projectors, sound recording equipment, animation and gaming software, and the ubiquitous PowerPoint - Instruction - formative assessment
Frondeville (2009)	- Use Quickwrites - short journal-writing assignment, summarize and predict an exam or quiz question. - Use signaling to allow everyone to answer a question - asking questions that allow for multiple answers or explanations - Introduce a presentation by having students pair up, talk to each other about their prior knowledge of the presentation, and generate a list of four questions.
Maheady & Gard (2010)	ClassWide Peer Tutoring (CWPT) is a low-cost, efficient instructional intervention that enhances student achievement by increasing opportunities for active responding and immediate feedback.
Rathvon (2008)	Response card instruction - students respond to teacher questions by writing on cards

Falcão et al., 2018.	▪ Fun and interactive context of strategy
Heflin et al. (2017)	▪ Use of mobile technologies for learning ▪ Promote collaborative learning
Goyal & Krishnamurthy (2018)	▪ Use of web-based system, for example, online learning material ▪ Better organize content ▪ Cater different type of students
Russell et al. (2016),	▪ Online quizzes ▪ Relevant to real-world applications
Wang (2017)	▪ Self-assessment and self-reflection activities ▪ Problem-solving activities
Dim, Kim & Orey (2017)	▪ Gamification approach for example gamified online discussion tool
Imlawi et al. (2015)	▪ Create online course-based social networks to interconnect with their students
Chen and Chiu (2016)	▪ Intergroup competition mechanism and integrate it into a multi-touch platform for collaborative design-based learning (DBL).

V. DISCUSSION AND RECOMMENDATIONS

Studying students' engagement is very challenging as engagement is heavily dependent on interaction, collaboration, and perception. From our study and analysis of the literature, we have identified several issues worth investigating.

A. Examining the impact of emotion on students' engagement

Despite the vast amount of research in students' engagement areas, students' emotional engagement still lacks conceptual clarity due to little attention from social studies' researchers (Pekrun et al., 2014; Skinner et al., 2008). The dominant focus has been on both the behavioral and cognitive subtypes given evidence for their pivotal role in influencing academic achievement. According to Kort et al. (2001) "Students who are anxious, angry, or depressed do not learn; people who are caught in these states do not take in information efficiently or deal with it well". Therefore, the importance of emotion in classroom teaching cannot and should not be taken lightly. This is verified by Pekrun et al. (2014), who surmise that research on goals' achievement and emotion have mostly been neglected as a factor in shaping students' engagement. Goals' achievement and emotion can be influenced by a number of factors including classroom goal structures, the quality of classroom instruction, autonomy support provided by teachers which have been shown to influence students' engagement (Pekrun, 2006; Klem & Connell, 2004).

A few studies have examined academic emotions as a predictor for dropout, in addition to their influence on achievement (Pekrun et al., 2014; Pekrun, et al., 2011; Respondek et al., 2017). It is recommended that investigation should be made to enhance emotional engagement in a classroom by examining the emotional states of a lecturer affected by the emotional states of students influenced by teaching strategies that subsequently impact on engagement level in a classroom.

Mazer (2013) found that teachers' immediate positive behaviors stimulate emotional arousal in students, which leads to greater emotional interest and engagement. The findings suggest that teachers' immediate response strongly predicted students' emotional interest than cognitive interest. Moreover, students who experience heightened emotional interest pay attention to a subject because they are energized, excited, and emotionally engaged by how the materials are presented (Mazer, 2013). The increase in emotional arousal enhances a student's attention, making it easier to receive more information and strengthen the engagement level.

B. Development of a configuration framework to generate an engagement-strategy for a particular classroom environment

As discussed in Section 4, there are a number of interesting techniques and strategies that have been proposed by researchers. However, we do not expect that teachers will review all these techniques and strategies to identify the suitable and applicable to their classroom environment. Therefore, we suggest here to develop a configuration framework that is able to generate a suitable strategy from some given inputs by a teacher such as a classroom environment, subject nature, available resources, teacher's capability and so forth. To exploit all the proposed techniques and strategies efficiently, they should be compiled in a system and by using the configuration framework, the system can offer some recommendations.

C. Utilizing the concept of computer-based simulation and virtual environment to study and investigate poor students' engagement instead of using traditional research tools

Many researchers have investigated the problems of poor engagement in classrooms using traditional methods such as the questionnaire survey and setting up experiments that consume considerable time, effort and with very limited settings. Furthermore, since these experiments are applied to humans, there are very limited settings to be tested and the

costs would be excessive if multiple settings are applied (Macal & North, 2009). On the other hand, modeling and simulation gain more attention (Mostafa et al. 2018, Ahmed et al. 2010A; Subramainan et al. 2016; Subramainan et al. 2017), the demand on reducing time and resource costs associated with development and validation of simulation models have also increased (Mostafa et al., 2013; Subramainan et al. 2015; Jassim et al. 2015). Simulation is useful when the phenomenon to be studied is not directly accessible or is difficult to observe directly (Davidsson, 2002; Mostafa et al., 2014; Ahmed et al. 2010B). The idea of experimenting with models rather on the real system is not something new (Mostafa et al., 2016; Mahmoud et al., 2018). For example, when an architect plans to investigate the behavior of a wind tunnel on a tower block model, they experiment on the model since the cost of experiment on a real tower block is too excessive. Thus, experimenting with simulation models as one way to obtain results and deriving the behavior of a model analytically is usually paramount because it provides information about how the model behaves given a range of inputs (Gilbert, 2008).

One of the potential methods that could be utilized is an agent-based social simulation (ABSS) (Mostafa et al., 2016; Mahmoud et al., 2016B). ABSS is a method to model systems that comprise of individual, autonomous, cooperating agents (Macal & North, 2009; Mahmoud et al., 2013). This method can be used to model human behaviors and their effects on others (Mahmoud et al., 2012). Social scientists have begun to convert social theories to computer programs (Mahmoud et al., 2016A). It is then possible to simulate social processes and carry out “experiments” that would otherwise be impossible (Mahmoud et al., 2015).

In general, ABSS is used to examine a phenomenon and solve problems that are challenging for humans to study and provide solutions. It can be used in a variety of areas (e.g., computer games) to represent human societies. Moreover, artificial and natural occurrences can be represented in the simulation. The reason for doing computer simulations is usually to gain a deeper understanding of a phenomenon. Research studies have proven that ABSS is a powerful tool for modeling and understanding phenomena in various areas such as economics and trading, health care, urban planning, and social events.

VI. CONCLUSION. LIMITATION, AND FUTURE WORKS

In this paper, existing research in various categories of students’ engagement is thoroughly reviewed. A collection of selected papers is classified into seven categories, overview; dimensions and indications; research studies; problems; factors influence engagement; methods to measure engagement; and techniques to improve engagement. We first describe the method to conduct this review, which includes search strategy, search terms, and selection process.

We then review the papers in each category, present the definitions and history of students’ engagement under the

overview category. Under the dimensions and indications category, we present the behavioral; emotional; cognitive and social dimensions and their indications. We then review 13 research studies on students’ engagement, identifying issues and challenges including disengaged issues in middle school, high school, and university levels. We found that the main challenge of studying the disengagement problem is that the traditional research tools lack flexibility and the technique is tedious and costly. Subsequently, we conduct a comprehensive analysis that includes data analysis on students’ engagement studies, number of selected articles by year of publication, rate of reviewed articles in different categories, number of articles in different categories by year of publication, and finally taxonomy of literature on students’ engagement. This is followed by a discussion and recommendations section. In which three potential research problems are discussed. The problems are, examining the impact of emotion on students’ engagement; development of a configuration framework to generate an engagement-strategy for a particular classroom environment; utilizing the concept of computer-based simulation and virtual environment to study and investigate poor engagement instead of using traditional research tools. The main limitation of this review is the number of reviewed papers in some categories, albeit the papers are reliable and are broadly representative collections. Another limitation is the rapid progress and development in this field has potential to limit the timeliness of the survey.

In our future work, we shall study and analyze the importance and challenges of emotions to students’ engagement to highlight factors influencing lecturers’ emotions and students’ emotional states that affect students’ engagement in a classroom. In addition, instead of using traditional methods in studying students’ engagement such as a questionnaire survey, we shall develop a simulation program that could potentially help researchers in the field of social studies to apply various environmental settings and study the impact of different strategies. This could be utilized to explore several issues such as, what are the strategies that can help to improve students’ engagement and which strategies are the most effective and which are the least effective; what is the optimal complexity level of a strategy to improve the engagement in shorter time; what is the average time to improve students’ engagement if it is low; and what are the ideal strategies a lecturer should consider at time 0 (before having evaluated the engagement level); and many other issues.

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